

CivicEye AI

AI-Powered Public Issue Reporting & Monitoring

Smart Digital Solution for Public Issue Management

YOUNG COMPUTER SCIENTIST COMPETITION 2026

National-Level School ICT Championship
Sri Lanka

AI ANALYSIS

AI-assisted processing of public issue reports.

SMART LOCATION

Location information supports accurate reporting.

MONITORING

Structured reporting and administrative review.

PROJECT INFORMATION

Project Name: CivicEye AI
Competition: Young Computer Scientist (YCS) 2026
Project Type: AI-Powered Web Application
Country: Sri Lanka

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PREMIUM PROJECT DOCUMENTATION

01

Project Overview

A centralized digital approach to public issue reporting

CivicEye AI is an AI-powered web application designed to help citizens report and monitor public issues in their communities. The platform provides a structured digital workflow for submitting information about problems such as road damage, garbage issues, water leakage and street-light problems.

Users can submit an image, select an issue type, provide a description and specify the location. The system then presents useful analysis information including confidence, priority, recommended authority, duplicate complaint status and report status.

02

The Problem

Challenges in conventional public issue reporting

01	Reporting Difficulty Citizens may not have a simple centralized platform for reporting different types of public issues.
02	Authority Identification Citizens may not know which authority is responsible for handling a particular public issue.
03	Repeated Complaints Multiple reports describing the same issue can make complaint organization more difficult.
04	Monitoring Challenges Public issue information can be difficult to organize for administrative review and follow-up.

03

CivicEye AI Solution

From citizen report to structured administrative information

CITIZEN REPORTING

Upload an image, select an issue, add a description and provide the location of the public problem.

AI-ASSISTED PROCESSING

Submitted information is processed and displayed as a structured analysis result for easier understanding.

ADMINISTRATIVE MONITORING

Administrators can review submitted reports, locations, priority, status and other relevant information.

ONE PLATFORM | ONE WORKFLOW | SMARTER MONITORING

Core Features

Key capabilities implemented in CivicEye AI

IMAGE-BASED REPORTING

Upload an image representing the public issue.

ISSUE DESCRIPTION

Provide detailed information about the reported problem.

SMART LOCATION

Enter or capture the location associated with a report.

AI ANALYSIS

Present AI-assisted analysis information for the report.

CONFIDENCE DISPLAY

Show the confidence associated with the analysis result.

PRIORITY DETECTION

Classify reports into Normal, Medium, High or Emergency.

AUTHORITY RECOMMENDATION

Recommend an appropriate authority based on the issue.

DUPLICATE DETECTION

Identify potentially repeated complaints.

ADMIN DASHBOARD

Provide administrators with a structured report view.

CITIZEN DASHBOARD

Provide citizens with access to submitted information.

STATUS TRACKING

Display the current state of submitted reports.

MULTILINGUAL INPUT

Designed to support English, Tamil and Sinhala interaction.

FEATURE WORKFLOW

Report → Analyse → Prioritize → Recommend Authority → Check Duplicate → Track Status → Admin Review

CivicEye AI combines citizen-facing reporting tools with administrative monitoring features, creating a practical foundation for organized public issue management.

Technology & Architecture

Technologies used to build the CivicEye AI application

FRONTEND	HTML5 CSS3 JavaScript Responsive web interface and interactive components.
BACKEND	Node.js Express.js Multer CORS dotenv Server-side report and upload handling.
DEPLOYMENT	Frontend: Web-based interface Backend: Railway

Project Structure

Main files and folders used by the project

FRONTEND index.html login.html citizen.html admin.html report.html ai-processing.html script.js style.css	BACKEND server.js package.json package-lock.json SUPPORT uploads/ Documentation/
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Backend API

Core endpoints used by the web application

GET /

Backend health check used to verify that the CivicEye AI backend is running.

POST /upload

Receives uploaded image and report information for processing.

GET /reports

Returns submitted reports for administrative monitoring.

DELETE /reports/:id

Removes a selected report from the backend.

Testing & Demonstration

Recommended evaluation workflow for judges

01

OPEN REPORTING PAGE

Open report.html and prepare a sample public issue report.

02

CREATE A REPORT

Upload an image, select an issue, add a description and provide the location.

03

SUBMIT REPORT

Submit the report and allow CivicEye AI to process the information.

04

VIEW AI ANALYSIS

Review issue, confidence, location, priority, authority, duplicate status and report status.

05

ADMIN REVIEW

Open admin.html to review the submitted report and its information.

Project Objectives

What CivicEye AI aims to demonstrate

- Encourage citizen participation in reporting public issues.
- Use AI-assisted processing to organize submitted information.
- Support public issue prioritization.
- Identify potentially duplicate complaints.
- Recommend a relevant authority.
- Provide digital administrative monitoring.

Future Improvements

Potential directions for further development

INTELLIGENCE

Advanced computer-vision issue detection
Improved multilingual AI support
Advanced analytics and prediction
Smarter issue classification

CONNECTIVITY

Real-time authority notifications
Interactive geographic issue maps
Cloud database integration
Integration with public-service authorities

Conclusion

A practical application of AI and web technologies

CivicEye AI demonstrates how AI and web technologies can be combined to address real-world public-service challenges. By bringing citizen reporting, image uploads, location information, AI-assisted analysis, priority detection, duplicate detection, authority recommendation and administrative monitoring together, the platform provides a foundation for a more organized public issue reporting process.

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